

A high-magnification scanning electron micrograph (SEM) of optic nerve tissue. The image shows a complex network of cellular structures, including myelinated axons, Schwann cell bodies, and various organelles. The texture is highly detailed, with many small, dark, circular features and elongated, fibrous structures. The overall appearance is a dense, intricate web of biological material.

Scanning Electron Microscopy (SEM)

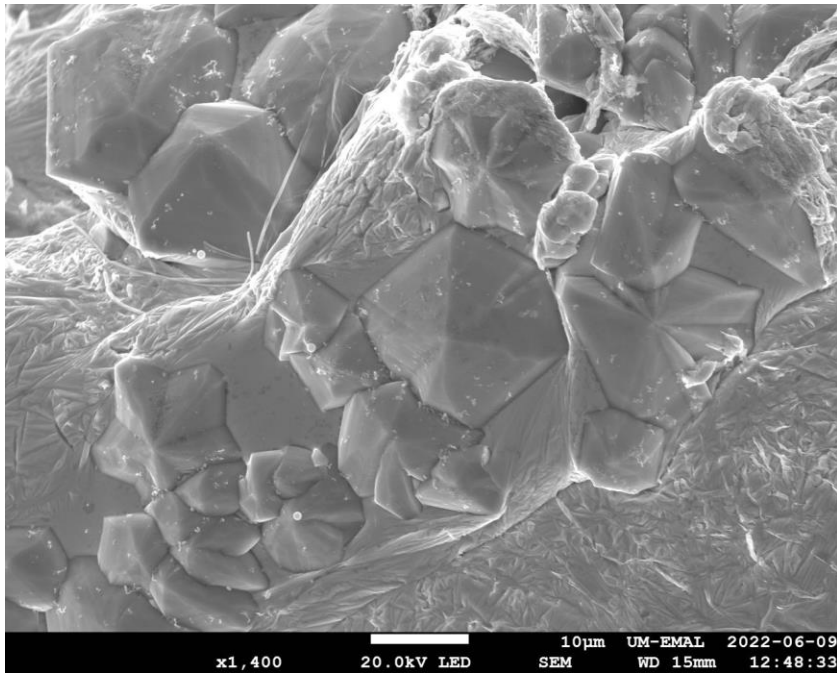
Suhama Saniz and Rachel Lee

Ph214

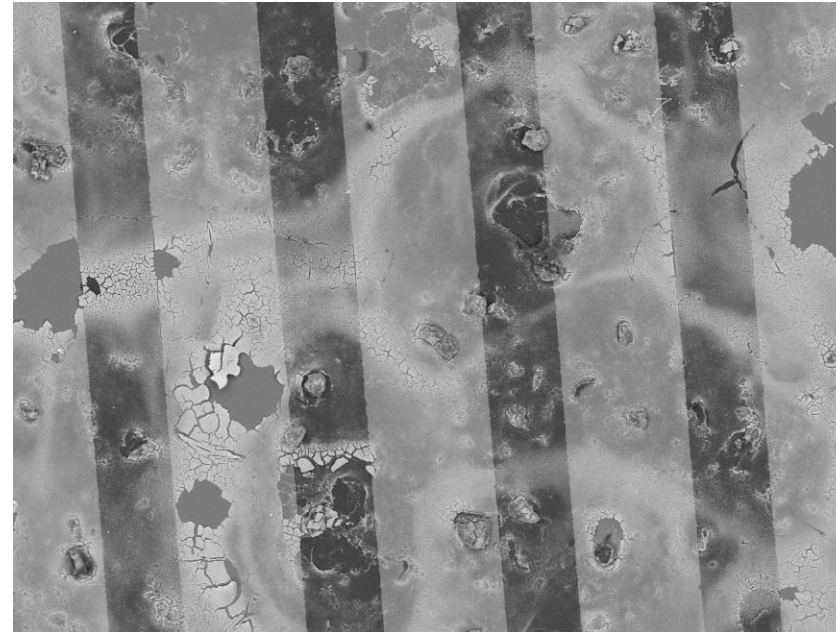
SEM image of optic nerve tissue. Courtesy of
Scanning Electron Microscopy | Nanoscience Instruments.
<https://www.nanoscience.com/techniques/scanning-electron-microscopy/>.

What is SEM?

- SEM – Scanning Electron Microscopy
- Scans a sample using a focused beam of electrons over some surface to create an image



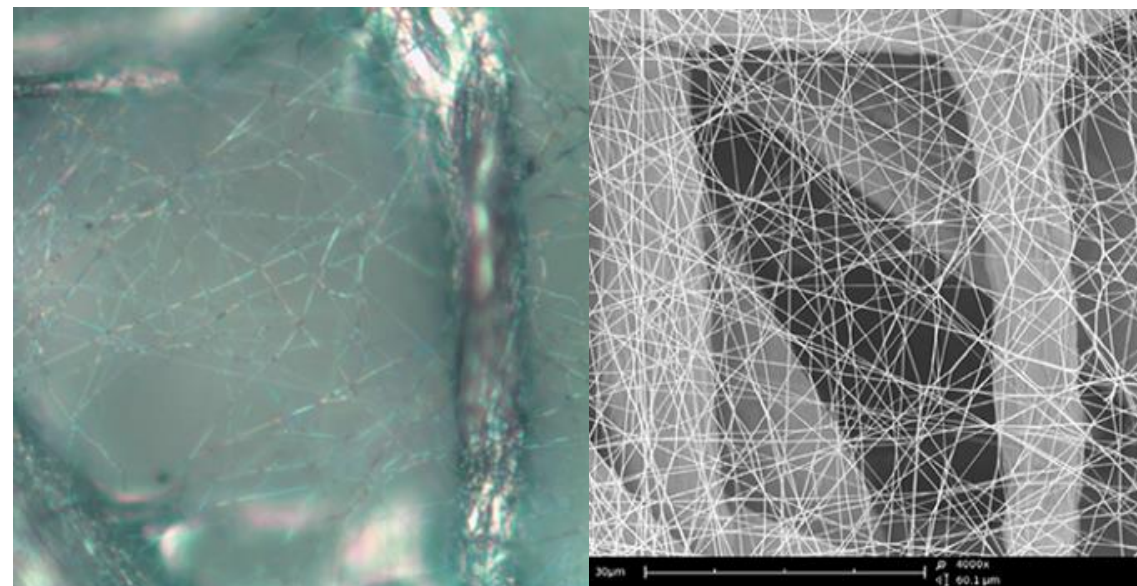
SEM image of Ge on HgIn alloy

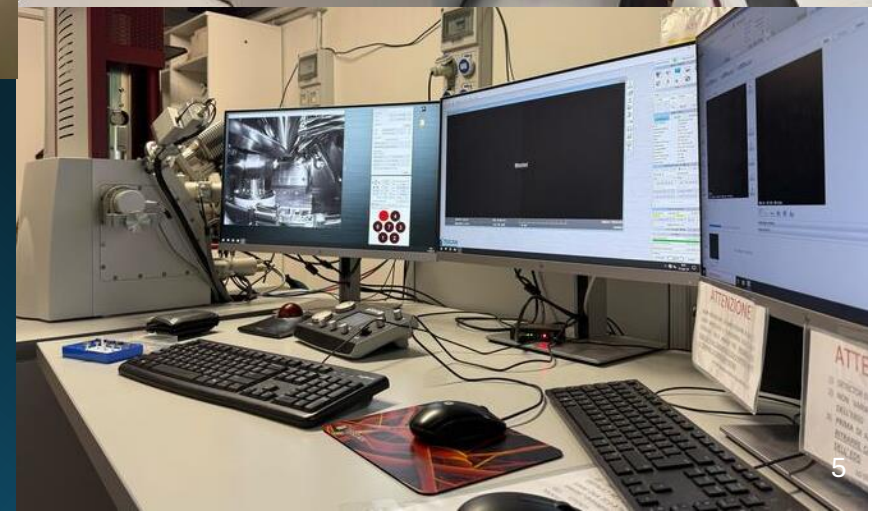
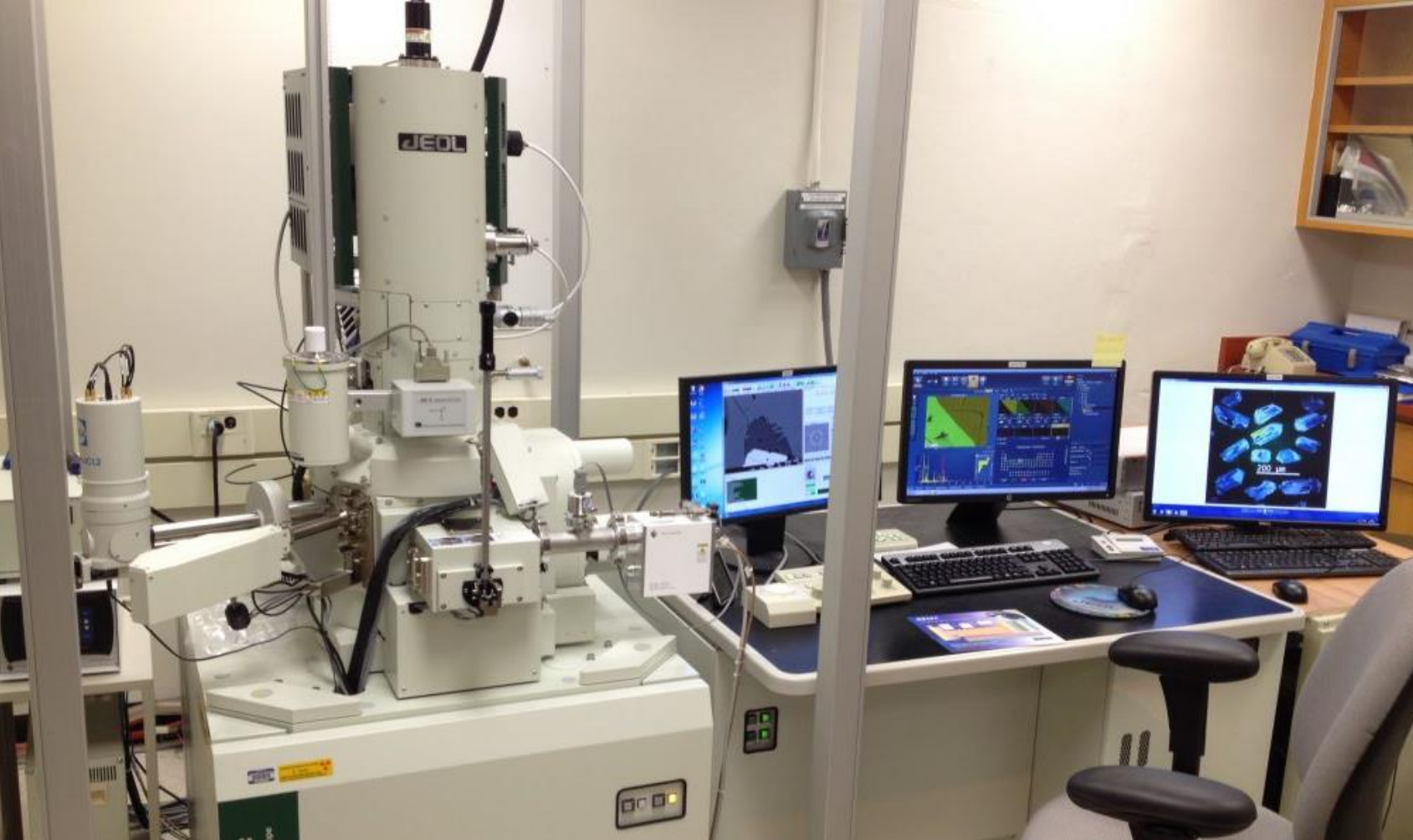


SEM image of CdS QDs layered
on a photoconductive device

Why is this a useful technique?

- High resolution down to the nanometer scale
 - Analyze the surface of a sample to greater detail
 - Observe materials unable to identify with the natural eye
 - Determine present elements in a given sample
 - Confirm crystallographic information of a sample
- Compared to an optical microscope:
 - Higher resolution (lower wavelengths, higher mag), more detail
 - Generally avoids particles in the air
 - Better contrast using electrons vs light

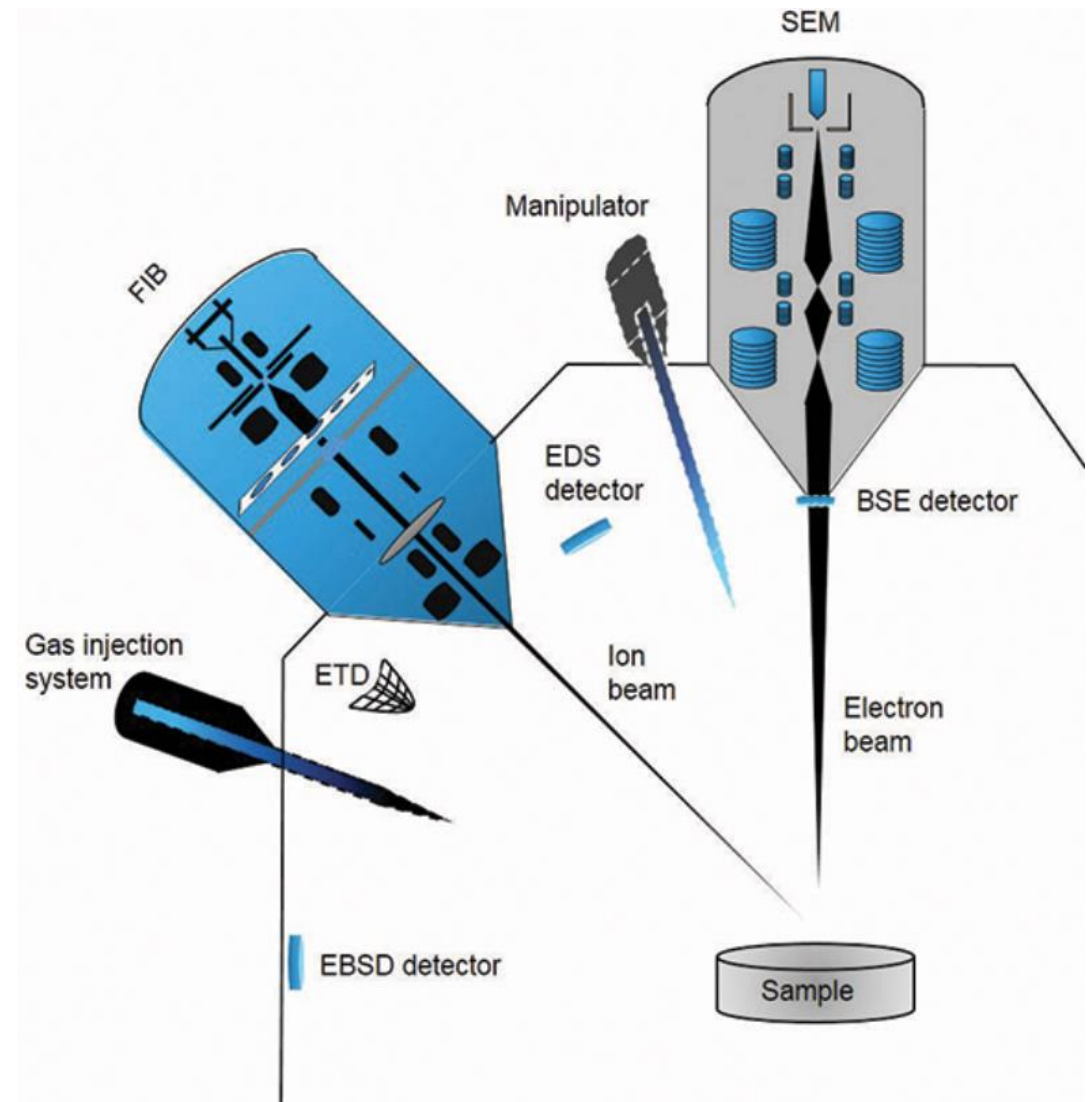




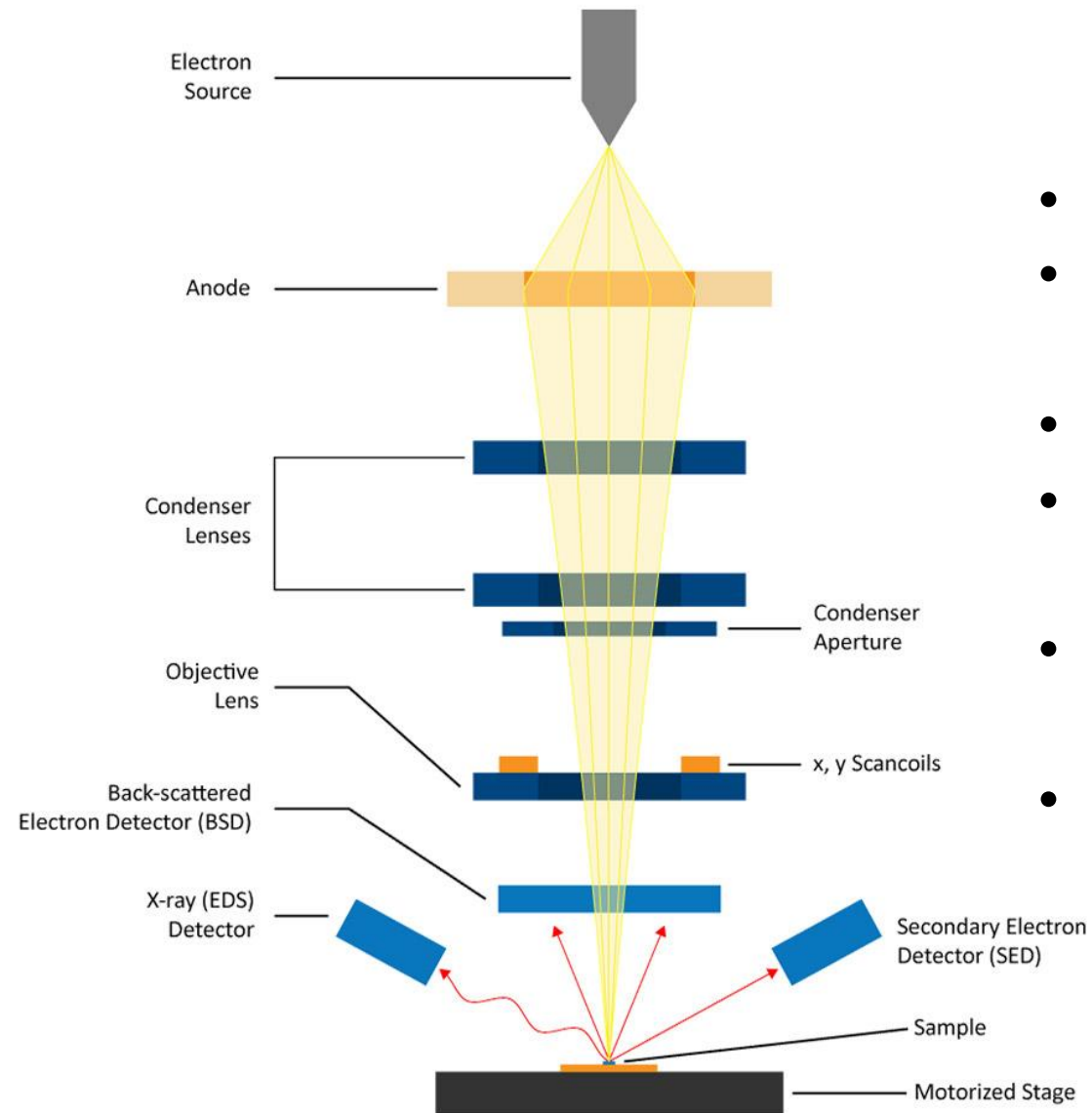
**Scanning Electron Microscope
(JEOL JSM-7800 FLV FE-SEM)**

Technical Setup

Schematic of a SEM Instrument



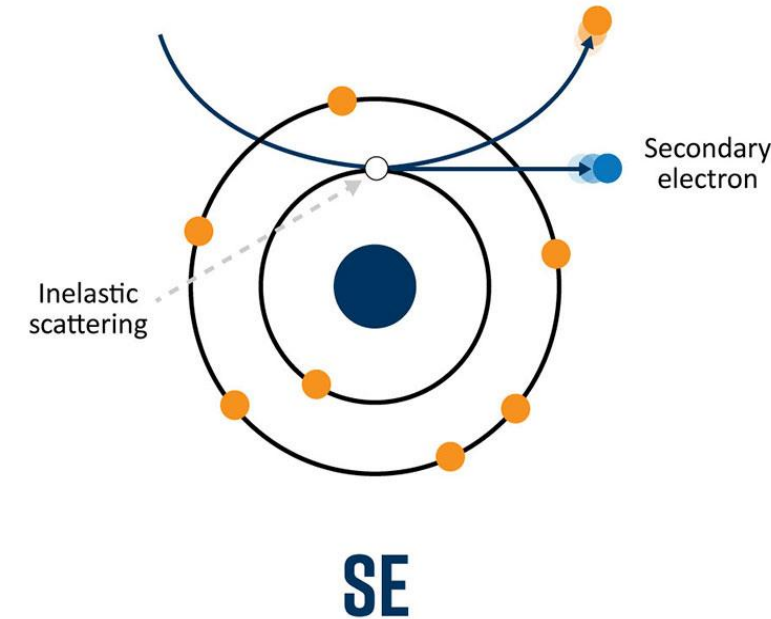
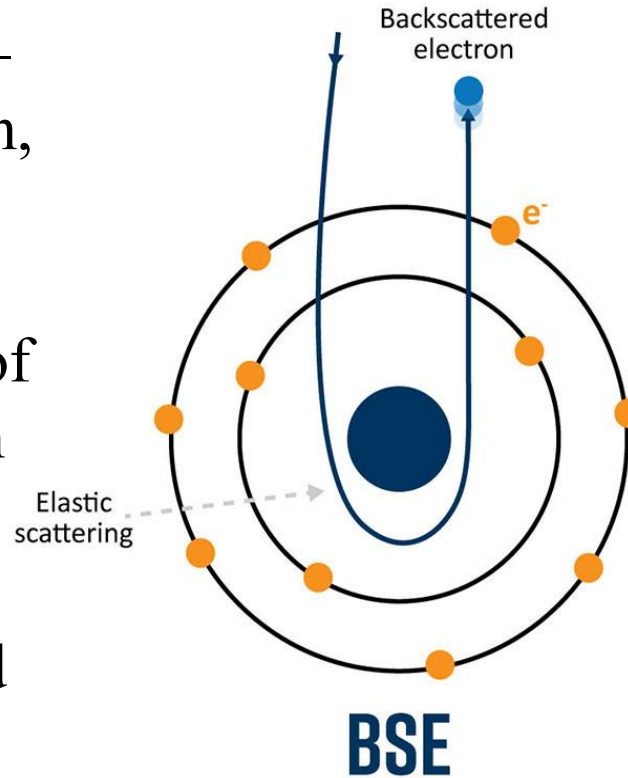
SEM Components



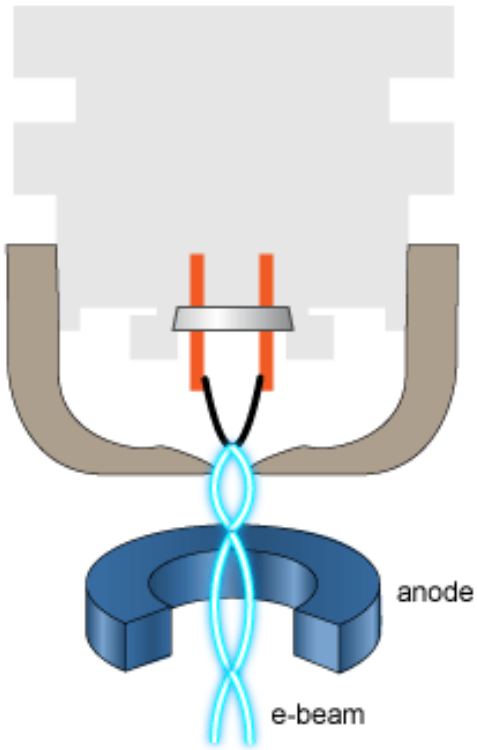
- **Electron Gun** – generates electrons
- **Anode, Lenses, and Apertures** – focus the electron beam, acceleration
- **Objective Lens** – focuses electron beam on sample
- **SE Detector** – collects secondary electrons from sample
- **BSE Detector** – collects backscattered electrons from sample
- **X-Ray Detector** – detect characteristic X-rays

Collecting the data

- **Backscattered Electrons (BSE)** – Provide compositional information, emitted from elastic scattering
- **Secondary Electrons (SE)** – Provide topographic information of the surface structure, emitted from inelastic scattering
- **X-rays** – Provide compositional information, emitted when ionized atoms relax to their ground state
 - e- beam displaces inner shell e-



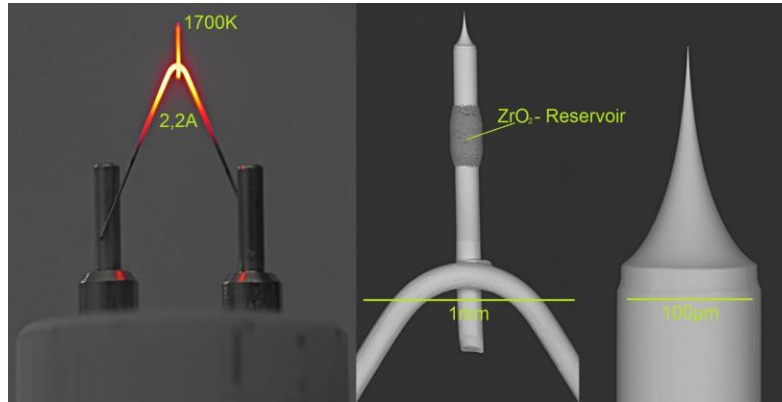
Electron Guns



Thermionic Gun

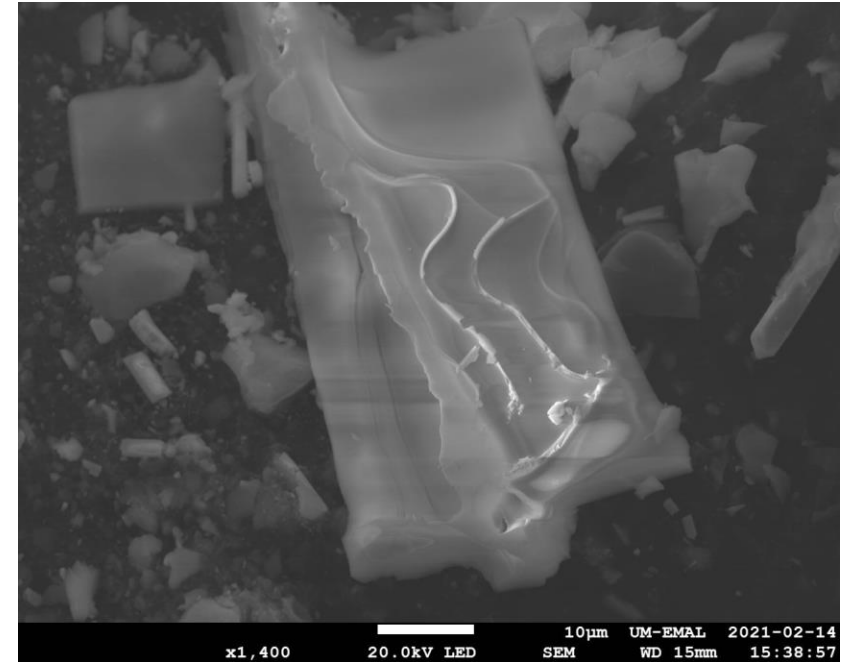
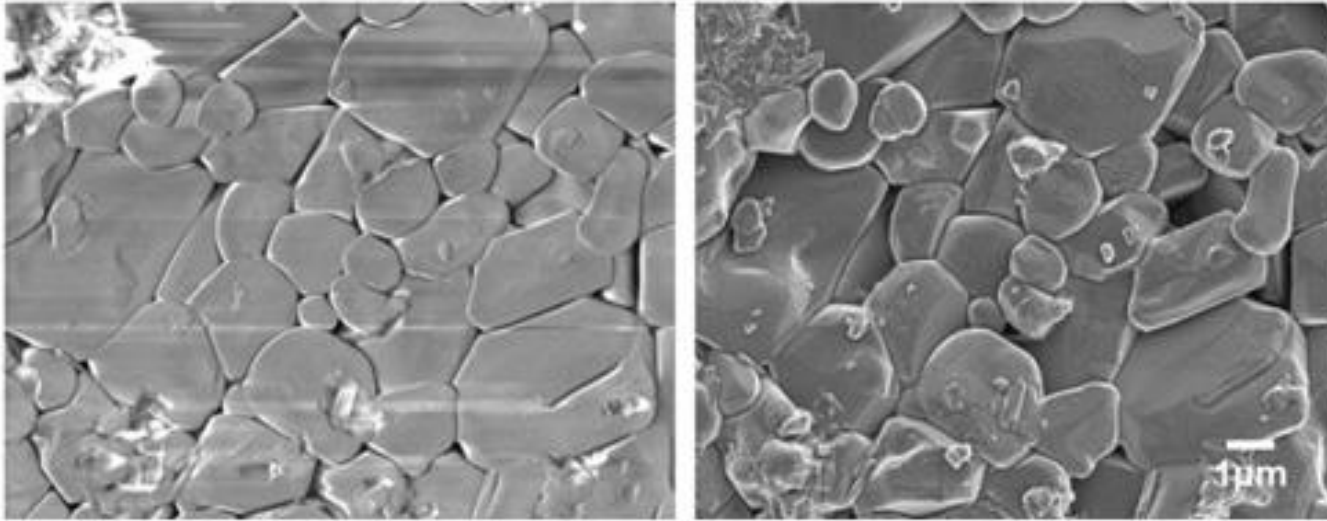


Field-Emission Gun



- Also known as thermal gun or a Schottky gun
 - **Schottky effect:** when an electric field is applied to a heated material, the potential barrier is lowered (reduction in **work function**) → e^- are more easily emitted
- Uses heated tungsten in an electric field to tunnel the potential barrier
 - Tungsten has a low work function (~ 4.5 eV), so popular material
- Useful for micro-area or nano-area analysis
- Uses a sharp tip tungsten emitter
- A strong electric field is concentrated on the tip of the emitter
- Smaller beam current and beam diameter, affords higher resolution

Limitations: Beware of Sample Charging



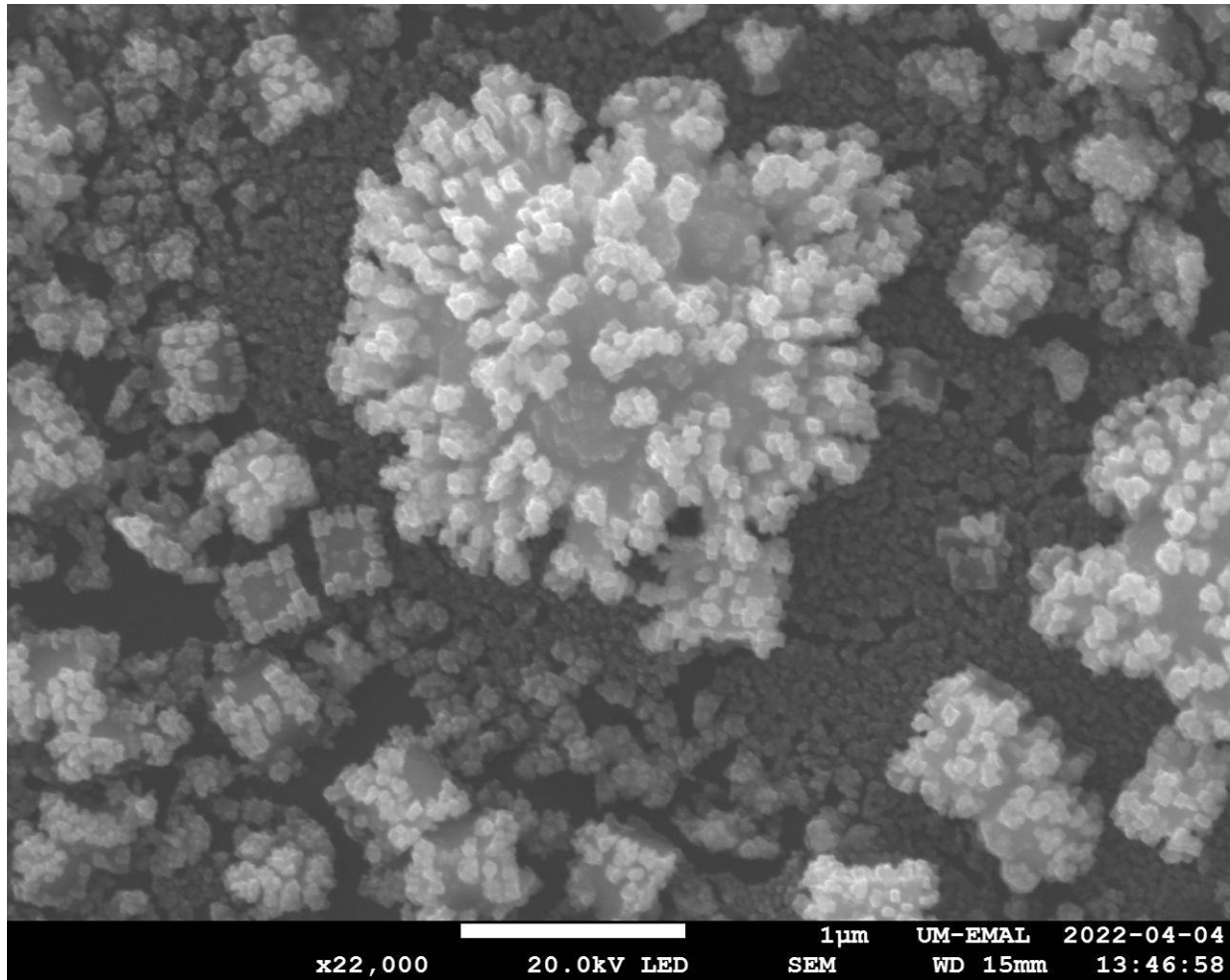
- A phenomenon that shows an anomalous-contrast in SEM imaging that generally occurs to samples that aren't conductive (accumulation of negative charge → abundance of signal detected → glowing areas!)
- What happens:
 - Contrast becomes higher (dark objects appear brighter)
 - Image distortion or drifting



One solution:
sputter sample
with gold first!

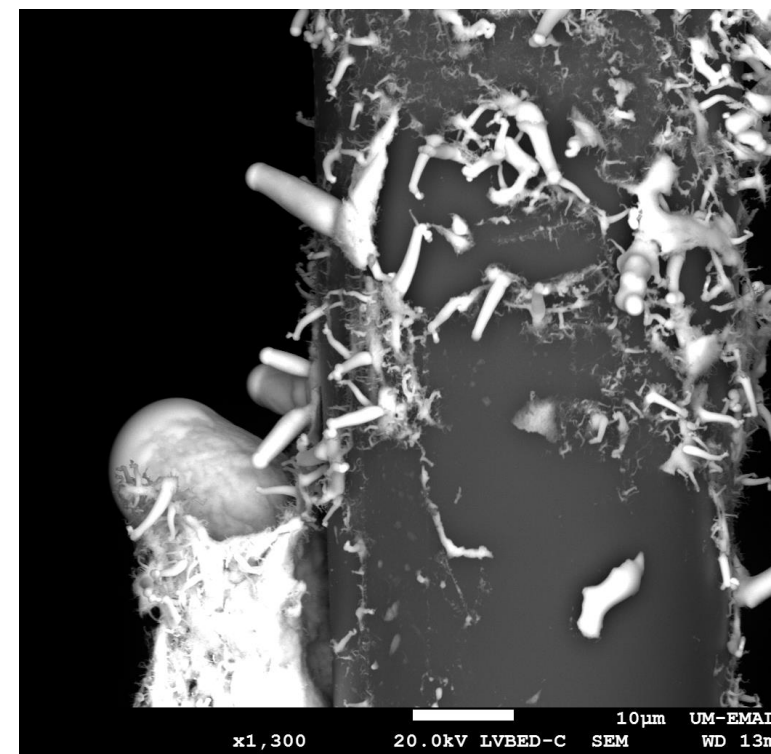
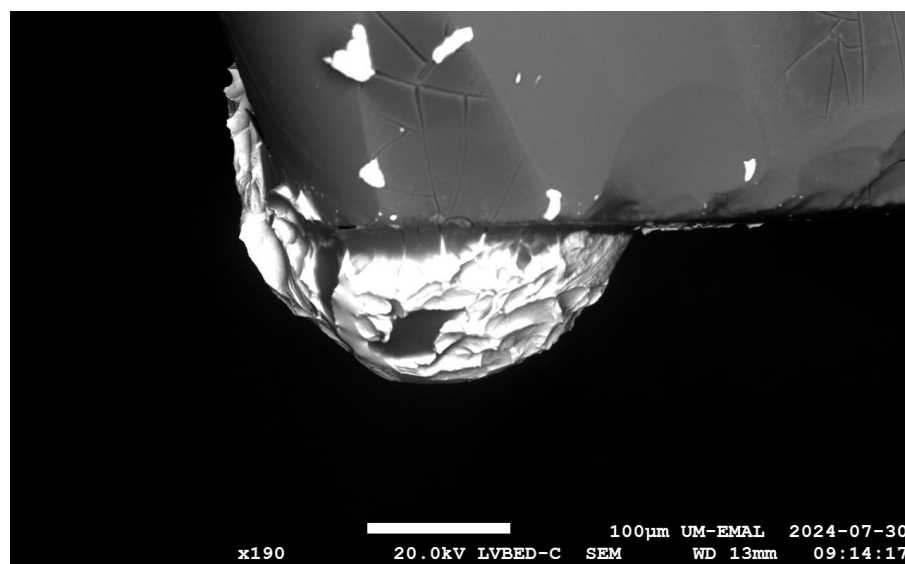
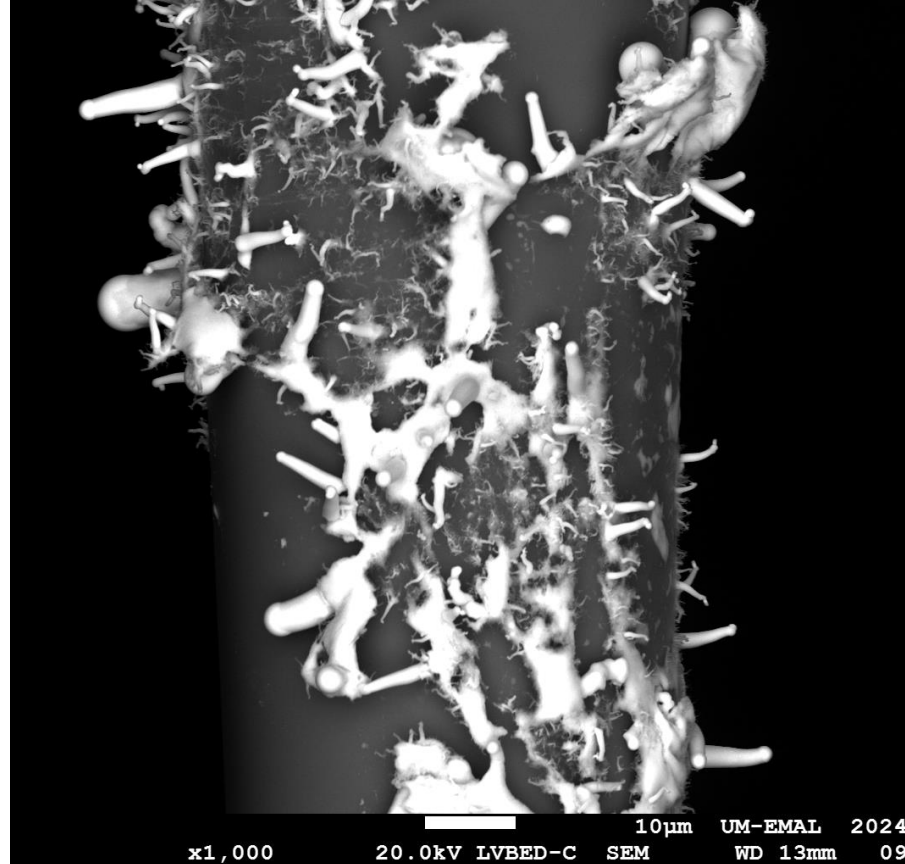
Applications

What do SEM images look like?



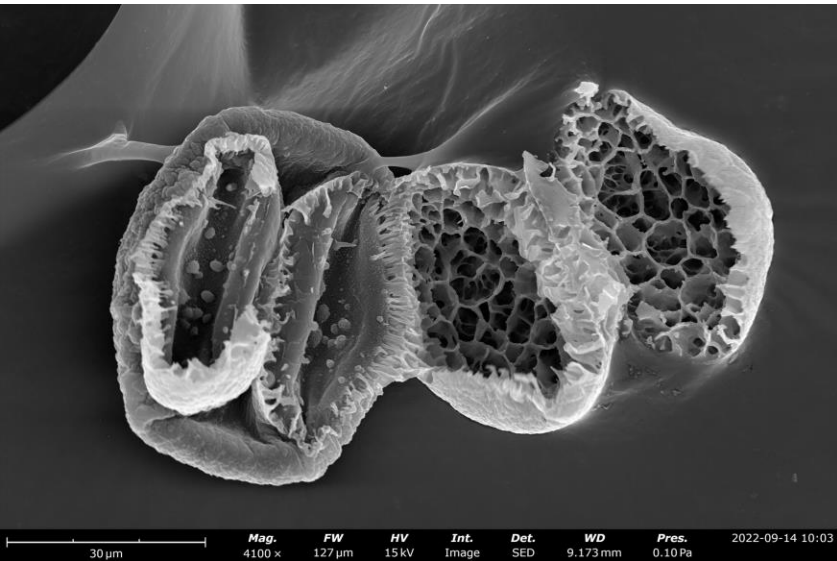
- Some parameters are recorded in the footer of the micrographs after imaging
- Depending on your settings and parameters, one can enhance the resolution and/or adjust focus/stigmatism to get better images
- Tips:
 - Higher accelerating voltage = higher resolution
 - Lower accelerating voltage affords more surface characterization (less penetration)

SEM images of Ge nanowires!

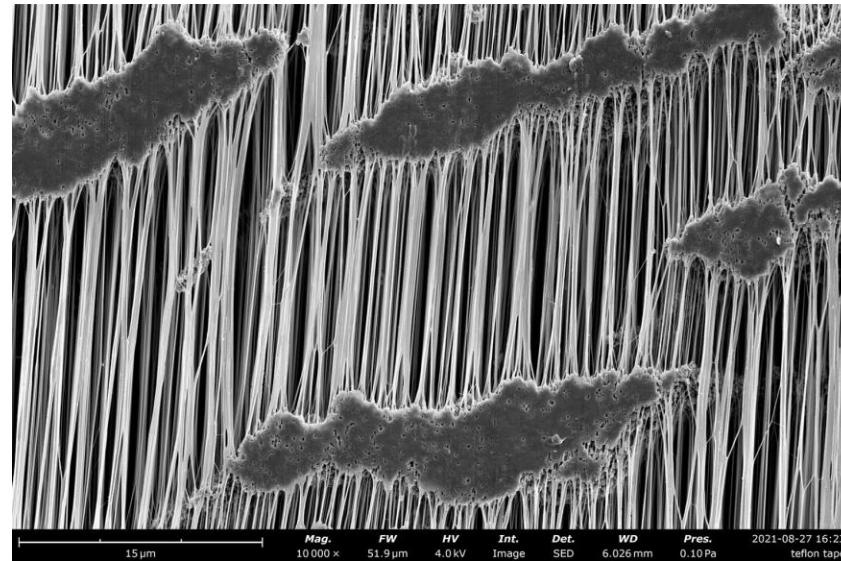


SE Images

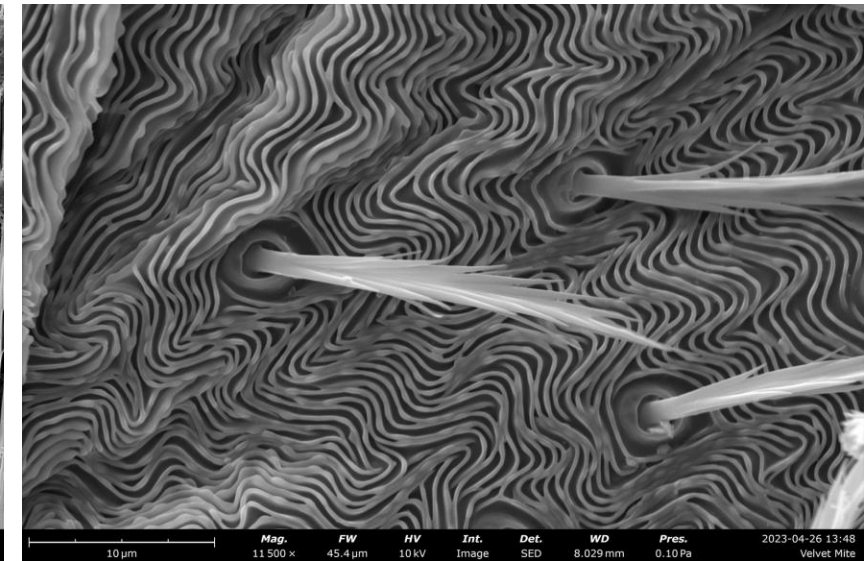
- Wonderful resolution and depth of field
- Resolution independent of sample material
- Mainly topological information



SE image of pollen

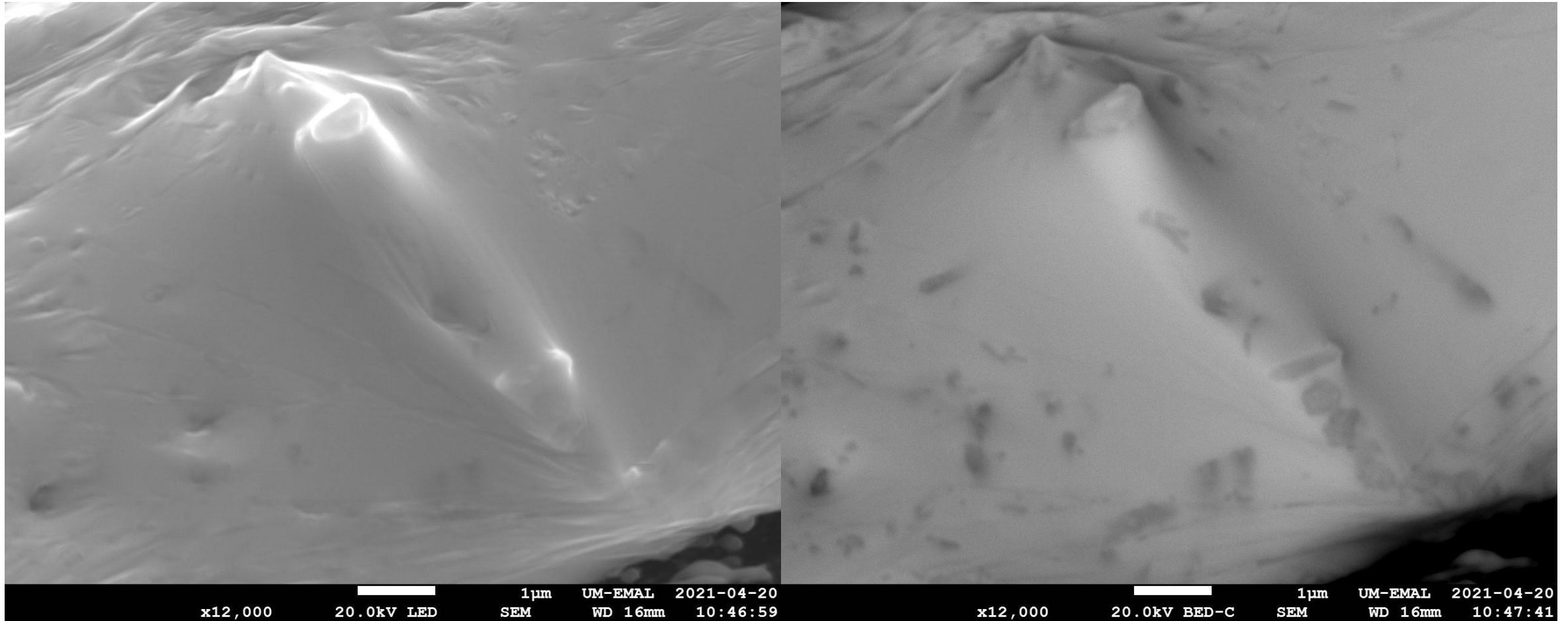


SE image of PTFE tape (aka Teflon tape, plumber's tape)



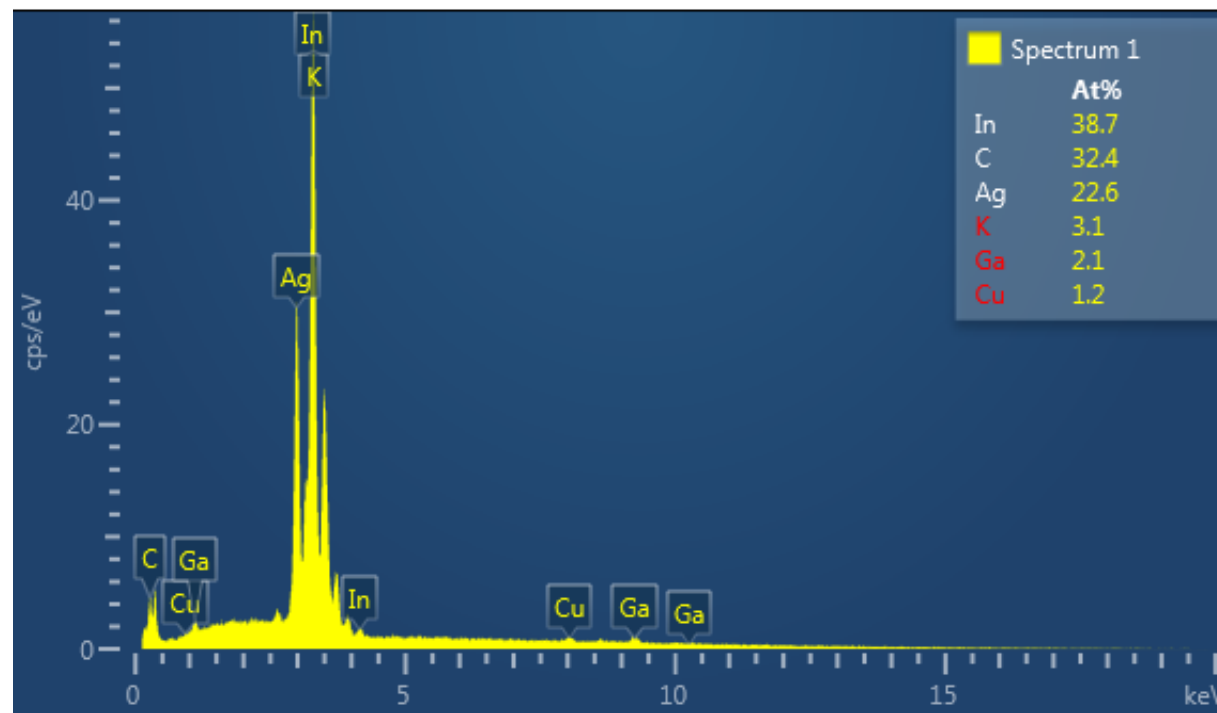
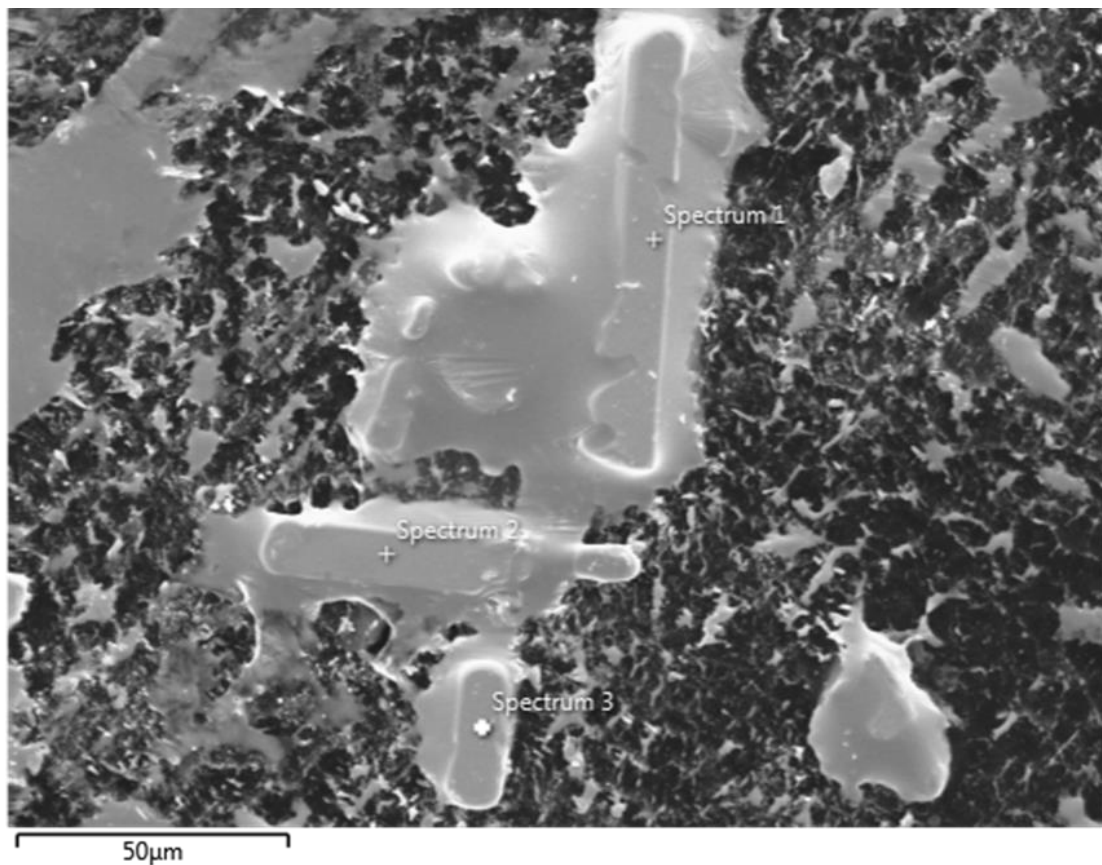
SE image of mite hairs

BSE Images



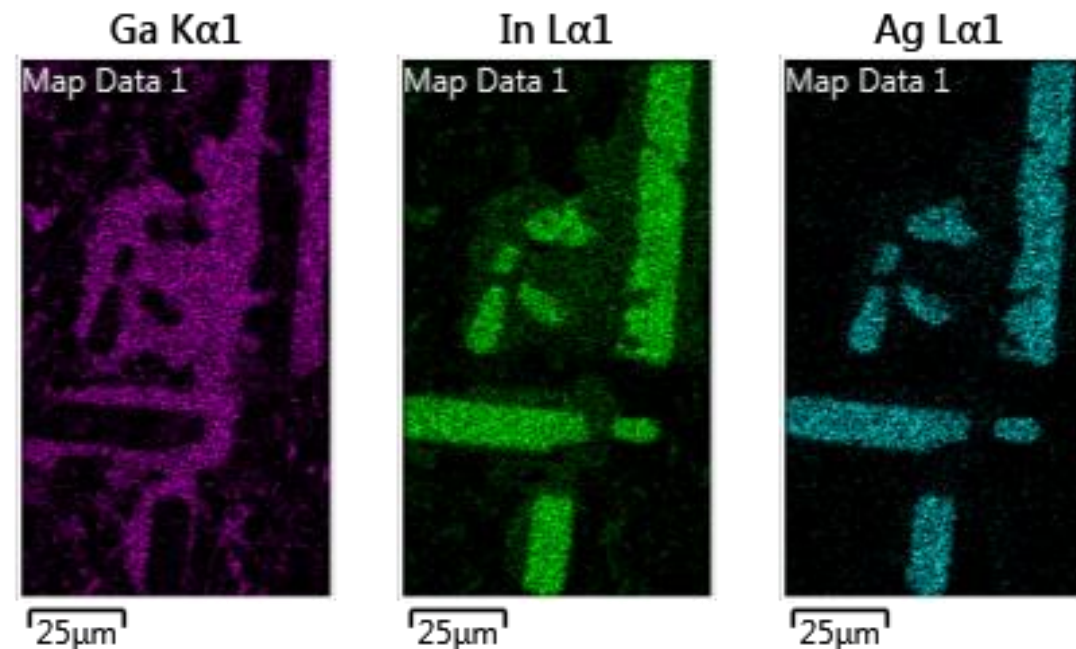
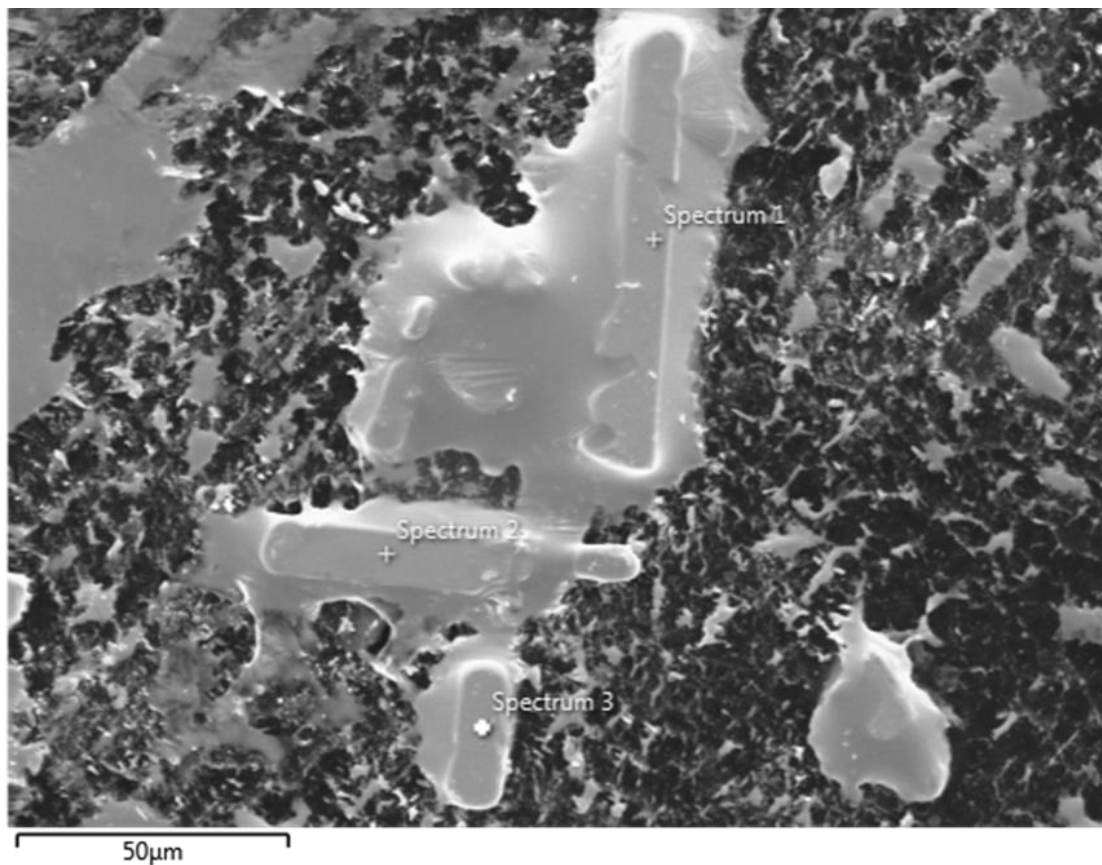
- More depth information but blurrier
- Can see details underneath the surface of the liquid metal
- In BSE, the higher the atomic number and heavier the element, the higher the contrast 18

Energy Dispersive X-ray Spectroscopy (EDX/EDS)



- Each element in the periodic table has unique energy level
- This technique can provide quantitative analysis on
 - Elements present in a given specimen
 - Concentration (at% or wt%) of an element in specimen

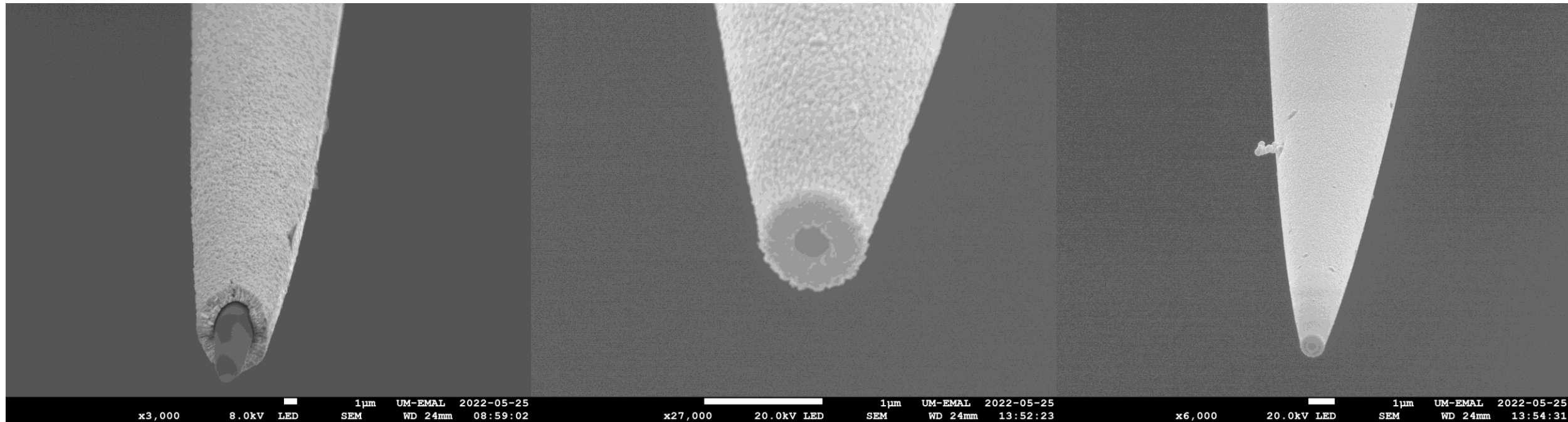
Energy Dispersive X-ray Spectroscopy (EDX/EDS)



- Elemental maps (among other modes like line scans) can be measured and analyzed

Summary

- SEM is a useful technique for imaging small or large samples and doing analysis of the elements of sample
 - Biomedical, material science, geological applications (to name a few!)
- A great method to get some preliminary observations of a sample before following up with other confirmation techniques



Some cool Au-coated glass pipette emitters

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